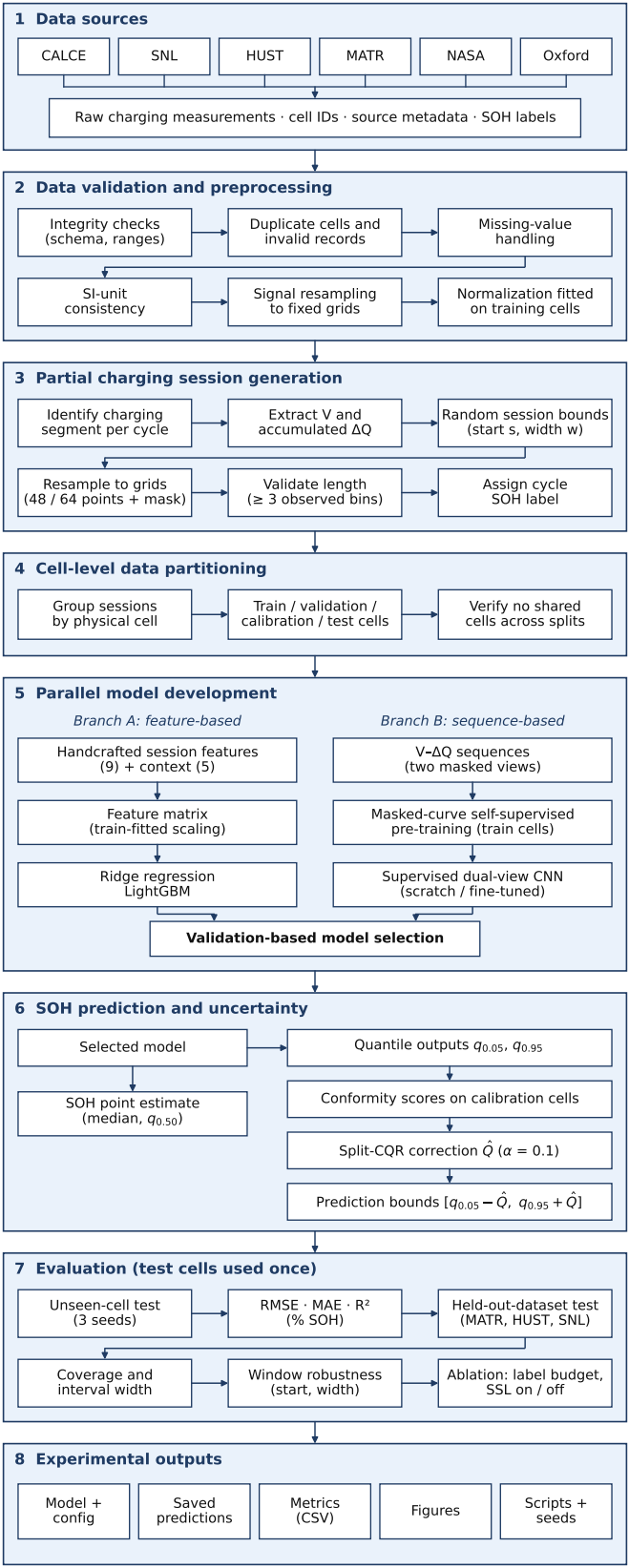
1 Data sources
CALCE
SNL
HUST
MATR
NASA
Oxford
Raw charging measurements · cell IDs · source metadata · SOH labels
2 Data validation and preprocessing
Integrity checks (schema, ranges)
Duplicate cells and invalid records
Missing-value handling
SI-unit consistency
Signal resampling to fixed grids
Normalization fitted on training cells
3 Partial charging session generation
Identify charging segment per cycle
Extract V and accumulated ΔQ
Random session bounds (start s, width w)
Resample to grids (48 / 64 points + mask)
Validate length (≥ 3 observed bins)
Assign cycle SOH label
4 Cell-level data partitioning
Group sessions by physical cell
Train / validation / calibration / test cells
Verify no shared cells across splits
5 Parallel model development
Branch A: feature-based
Handcrafted session features (9) + context (5)
Feature matrix (train-fitted scaling)
Ridge regression LightGBM
Branch B: sequence-based
V–ΔQ sequences (two masked views)
Masked-curve self-supervised pre-training (train cells)
Supervised dual-view CNN (scratch / fine-tuned)
Validation-based model selection
6 SOH prediction and uncertainty
Selected model
SOH point estimate (median, $q_{0.50}$)
Quantile outputs $q_{0.05}$, $q_{0.95}$
Conformity scores on calibration cells
Split-CQR correction $\hat{Q}$ ($\alpha = 0.1$)
Prediction bounds $[q_{0.05} - \hat{Q},\ q_{0.95} + \hat{Q}]$
7 Evaluation (test cells used once)
Unseen-cell test (3 seeds)
RMSE · MAE · R² (% SOH)
Held-out-dataset test (MATR, HUST, SNL)
Coverage and interval width
Window robustness (start, width)
Ablation: label budget, SSL on / off
8 Experimental outputs
Model + config
Saved predictions
Metrics (CSV)
Figures
Scripts + seeds